

Exercise

1. What are the 7 basic principles of object orientation? Provide a brief description for each principle.

- Divide-and-Conquer - Divide the overall program into a number of objects that will interact with each other to solve the problem
- Encapsulation and Modularity - Decide the attributes and actions for each object. The goal of each object is a self-contained module with a clear responsibility and necessary attributes and actions
- Public Interface - Clarify how objects should interact with each other using a clear interface
- Information Hiding - Hide certain details of design and performance from other objects in order to work together cooperatively. Each object should shield its users from unnecessary details of how it performs its role
- Generality - Design each object for a particular kind of task which involves the uses of software libraries. In other words, objects should be designed to be as general as possible
- Extensibility - Objects should be designed so that they can be extended to handle new tasks.
- Abstraction - Ability to focus on the important features of an object when trying to work with large amounts of information. Objects can ignore many of the attributes and focus only on important attributes.

2. State the benefits of OOP

- Save development time and costs by reusing code
- Easier debugging as classes can be tested independently and reused objects that have already been tested